HA THU **TRAN** | BACKEND ENGINEER @ TIKTOK

SINGAPORE CITIZEN, H1B1 US VISA ELIGIBLE | TRANHATHU.COM • TRANHATHUWORK@GMAIL.COM • LINKEDIN.COM/IN/TRAN-HA-THU

Hello! Due to a recent re-organization that significantly changed my job scope, I am looking for a new role that lets me build cool products again. I love engineering and improving the reliability of scalable software, and exploring how the latest tech and AI expedite workflows and development.

EXPERIENCE

TikTok • Backend and Site Reliability Engineer **Jul 2026 – Present**

- Improved service reliability, observability and alarm handling flow for 12 teams (200 unique users) through development of internal apps.
- Maintained SLOs and service reliability by debugging and solving around 50 service issues a week as an on-call engineer.
- Saved close to $5 million by coordinating the allocation of computing resources with multiple teams in both English and Chinese.

TikTok • Software Engineering Intern **Jan 2026 – Mar 2026**

- Developed internal products with TypeScript and Go to improve the reliability and observability of our internal services.

GIC • Software Engineering Intern **Jun 2025 – Sep 2025**

- Optimized workflows for over 300 staff by engineering products for public market investments using TypeScript, C#, Python and MSQL.

National University of Singapore • Teaching Assistant **Aug 2023 – Dec 2025**

- Taught over 1000 students programming concepts, JavaScript, Java and Python to students through self-prepared tutorial sessions.
- Rated 4.8/5.0 (CS1101S) and 4.7/5.0 (CS2103T) in the student feedback exercise (Department average: 4.3).

Uptale • Software Engineering & AI Intern **Jan 2025 – Jun 2025**

- Revolutionized the creation of gamified educational experiences for over 100 customers through the implementation of a retrieval-augmented generation (RAG) pipeline with TypeScript, C# and Azure RAG that could create complete 3D Virtual Reality scenarios with AI.

Housing Development Board Singapore • Data Science Intern **May 2024 – Aug 2024**

- Minimized manual labor by training a model to predict commercial rent prices with an average of 88% accuracy and drove business processes with Python and machine learning, using a variety of models including Random Forest, XGBoost and K Nearest Neighbors.
- Improved efficiency by 80% through the automation of financial report generation using Excel and Excel VBA.

Eastspring Investments • Data Engineering Intern **May 2023 – Aug 2023**

- Maintained over 100 production and raw tables for the company's database using Python and SQL.
- Optimized and developed an Artificial Intelligence chatbot using ChatGPT's API.

ASTAR Research • Artificial Intelligence Intern **Dec 2019 – Jan 2020**

- Trained a working robot arm capable of clearing 10 types of trash from scratch using Python and OpenCV as a 16-year-old.

EDUCATION

Bachelor of Computing – Computer Science (First Class Honors) **Aug 2022 – May 2026**

National University of Singapore (NUS)

GPA: 4.6 / 5.0 – **First Class Honours, Distinction in Software Engineering, Dean's List**

- Founding engineer for a Healthtech startup, which attained 2nd @ NUS Social Impact Catalyst and top 10 for TigerLaunch Pitch.
- Planned over 20 different university events as a residential assistant and leader in various CCAs and committees.
- Attained 2nd for a SMU Quantum Computing Hackathon, the SOC Student Bronze Award and the Tembusu College best project award

SKILLS & TECH STACKS

- **Skills:** Agentic Development • Full-Stack Development • Scalable Systems • CI/CD • Version Control • Deployment
- **Programming Languages:** Python • Go • C# • TypeScript • Java
- **Frontend:** ReactJS • Tailwind CSS • Cypress for testing • HTML • CSS
- **Backend & Databases:** PostgreSQL • MySQL • Microsoft SQL Server • CosmosDB

PROJECTS

My personal AI agent – an OpenClaw setup: https://www.tranhathu.com/projects/openclaw_agent/

- Leveraged LanceDBPro, Telegram bots to create a personal agent with an in-depth knowledge of me and my preferences.
- Having fun exploring different tools, such as Playwright MCP for browser access, and JEV/OpenAI decisions API to expedite workflows.

Call Analyzer – a self-designed Chinese learning system: https://www.tranhathu.com/projects/call_analyzer_chinese_learning/

- Started from my own desire to improve my Chinese abilities. Built with PostgreSQL, Wails (Go), React and OpenRouter API, and deployed on GCP to automatically capture new vocabulary from lessons and daily content, and provide active recall through a Telegram bot.
- Improved personal learning efficiency by over 100%.

Trader's Edge – a game to teach trading (Final Year Project): https://www.tranhathu.com/projects/traders_edge/

- A gamified investment training platform that teaches players basic trading techniques and allows classrooms to play together.
- Built with Python, React, and Yahoo Finance API to simulate order books and world events to closely mimic real-world market movement.

Personal Portfolio Website: https://www.tranhathu.com/

- Used React, HTML, CSS, *and* JavaScript to create a website that showcases all my work and experiences.

Demo Forum Website – Self-taught, hand-coded full stack project before the age of AI: https://www.tranhathu.com/projects/foroom/

- Self-learned to manually build a full-stack forum website as a 1st year student. Used React, pure CSS, Go backend and SQLite.